1. (a) $\bar{x} = \frac{154}{20} = 7.7$ (1)

$$sd = \sqrt{\frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right]}$$

$$= \sqrt{\frac{1}{19}\left[1568.94 - \frac{154^2}{20}\right]}$$

$$= \sqrt{20.1653}$$

$$= 4.49$$ (1)

(b) Median is at $x_{(10.5)} = 7.2$ (1)

Q_1 is $x_{(1+19\times0.25)} = x_{(5.75)}$

$x_{(5)} = 5.8$, $x_{(6)} = 6.0 \Rightarrow x_{(5.75)} = 5.8 + 0.75 \times 0.2$

$= 5.95$ (1)

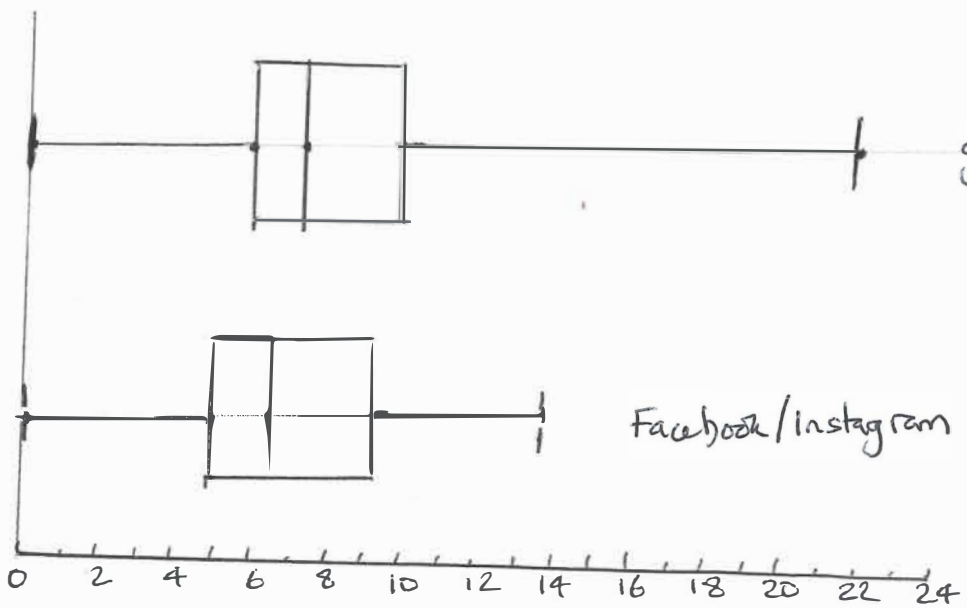
Q_1 is at $x_{(1+19\times0.75)} = x_{(15.25)}$.

$x_{(15)} = 10.2$, $x_{(16)} = 10.4 \Rightarrow x_{(15.25)} = 10.2 + 0.25 \times 0.2$

$= 10.25$.

So IQR = 4.3 (1)

(c)



(1) (1)

(c) ctd.

It does look as if people are favouring gaming over Facebook as the median is higher. Note, though, that the spread is more for gaming with a possible outlier. (1)

2. (a)

(i) $$\begin{aligned} \text{MSE}(g(\underline{X})) &= E((g(\underline{X})-\theta)^2) \\ &= E((g(\underline{X}))^2) - 2E(g(\underline{X}))\theta + \theta^2 \\ &= [E((g(\underline{X}))^2) - (E(g(\underline{X})))^2] + [(E(g(\underline{X})))^2 - 2E(g(\underline{X}))\theta + \theta^2] \\ &= \text{Var}(g(\underline{X})) + [E(g(\underline{X})) - \theta]^2 \\ &= \text{Var}(g(\underline{X})) + (\text{bias}(g(\underline{X})))^2. \end{aligned}$$ (1) (1)

((ii) It helps calculate the MSE for a particular example as it is usually easy to obtain bias and var. 1

(b) (i) $E(\hat{\lambda}_1) = E\left(\frac{X_1+X_2}{2}\right) = \frac{1}{2}E(X_1) + \frac{1}{2}E(X_2)$

$= \frac{1}{2}\lambda + \frac{1}{2}\lambda = \lambda$

So bias $(\hat{\lambda}_1) = \lambda - \lambda = 0$ (1)

$\text{Var}(\hat{\lambda}_1) = \frac{1}{4}\text{var}(X_1) + \frac{1}{4}\text{var}(X_2) = \frac{1}{2}\lambda$. (1)

bias $(\hat{\lambda}_2) = E\left(\frac{\Sigma x}{n}\right) - \lambda = \Sigma \frac{1}{n} E(x) = \lambda - \lambda = 0$ (1)

$\text{Var}(\hat{\lambda}_2) = \text{var}(\bar{x}) = \frac{1}{n^2}\Sigma \text{var}(x) = \frac{\lambda}{n}$ (1)

(ii) $\text{MSE}(\hat{\lambda}_1) = \text{var}(\hat{\lambda}_1) = \frac{1}{2}\lambda$ (1)

$\text{MSE}(\hat{\lambda}_2) = \frac{\lambda}{n}$ (1)

Since for $n > 2$, $\text{MSE}(\hat{\lambda}_2) < \text{MSE}(\hat{\lambda}_1)$ $\hat{\lambda}_2$ is the more efficient estimator. (1)

(iii) $\mu(\lambda) = \lambda$ (population) (1)

$\bar{X}$ is the sample mean.

Equate the two $\Rightarrow$ $\hat{\lambda} = \bar{X}$ (1)

3 a).

1. Contribution to $L(\beta)$ over the 10 day monitoring period is given by

$$L(\beta) = \beta e^{-\beta x_1} \cdot \beta e^{-\beta x_2} \beta e^{-\beta x_3} \quad (1)$$

$$= \beta^3 e^{-\beta(x_1+x_2+x_3)}$$

$$= \beta^3 e^{-16\beta} \quad (1)$$

2. Contribution to $L(\beta)$ for every patient who does not have an attack before observation period ends.

$$1 - \int_0^{10} \beta e^{-\beta x} dx$$

$$= 1 - \left\{ \left[-e^{-\beta x} \right]_0^{10} \right\}$$

$$= 1 - \left\{ -e^{-10\beta} + 1 \right\}$$

$$= e^{-10\beta}$$

There are 6 people in this category so their contribution is

$$L(\beta) = (e^{-10\beta})^6 = e^{-60\beta}. \quad (1)$$

$\therefore$ Putting 1 and 2 together gives

$$L(\beta) = \beta^3 e^{-16\beta} e^{-60\beta} = \beta^3 e^{-76\beta} \quad (1)$$

(b) $l = \log(L) = \log \beta^3 - 76\beta$

$= 3\log\beta - 76\beta$ (1)

$\frac{dl}{d\beta} = \frac{3}{\beta} - 76$ (1)

$\left(\frac{d^2l}{d\beta^2} = -\frac{3}{\beta^2}\right.$ which is negative) *

Equate $\frac{dl}{d\beta}$ to 0 $\Rightarrow \frac{3}{\hat{\beta}} = 76 \Rightarrow \hat{\beta} = 0.0395$ (1)

This is maximum because of (*) (1)

(c) It might take a long time for some patients to develop heartburn so it would be a poor use of resources to monitor for much longer . 1

4(a) We can use normal approximation since the sample is large.

C.I. $= 4.2 \pm 2.5758 \times \frac{5}{\sqrt{n}}$ (1)

$= 4.2 \pm 2.5758 \times \frac{1.6}{\sqrt{50}}$

$= 4.2 \pm 0.583$

$= (3.617, 4.783)$ (1)

(c) The assumption is that because we are dealing with large samples, by the CLT the differences of means follow a normal distribution. (1)

(b) $H_0 : \mu_1 - \mu_2 = 0$ (1)

$H_1 : \mu_1 - \mu_2 \neq 0$

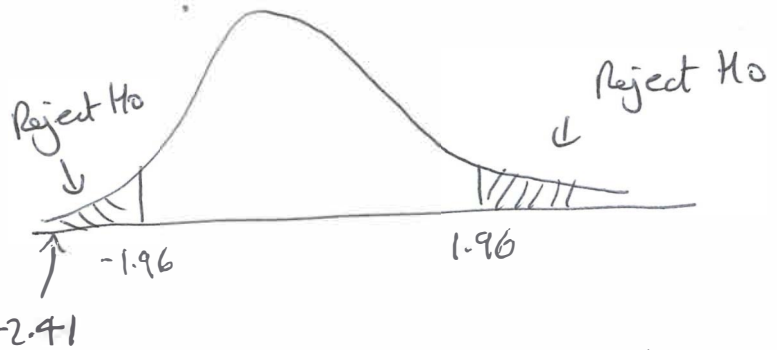
Large samples so test statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$ (1)

$$= \frac{4.2 - 4.9}{\sqrt{\frac{1.6^2}{50} + \frac{1.4^2}{60}}}$$

$$= \frac{4.2 - 4.9}{0.290}$$ (1)

$$= -2.41$$ (1)



At the 5% level there is evidence to reject Ho. The mean number of units of alcohol consumed per week is not the same between age groups. (1)

5.

(a) $p(\text{Type I error}) = p(\text{Reject } H_0 \text{ when it is true}).$

$= p(x \geq 7 \mid p = 0.5)$ (1)

This is a binomial distribution with $n = 10$. (1)

Tables give $p(x \leq 6) = 0.8281$.

$\Rightarrow p(x \geq 7) = 1 - 0.8281 = \underline{0.1719}$ (1)

(b) $p(\text{Type II error}) = p(\text{Accept } H_0 \text{ when } H_1 \text{ is true}).$

(i) $p(x \leq 6 \mid p = 0.7) = \underline{0.3504}$ (2)

(ii) $p(x \leq 6 \mid p = 0.8) = \underline{0.1209}$ (2)

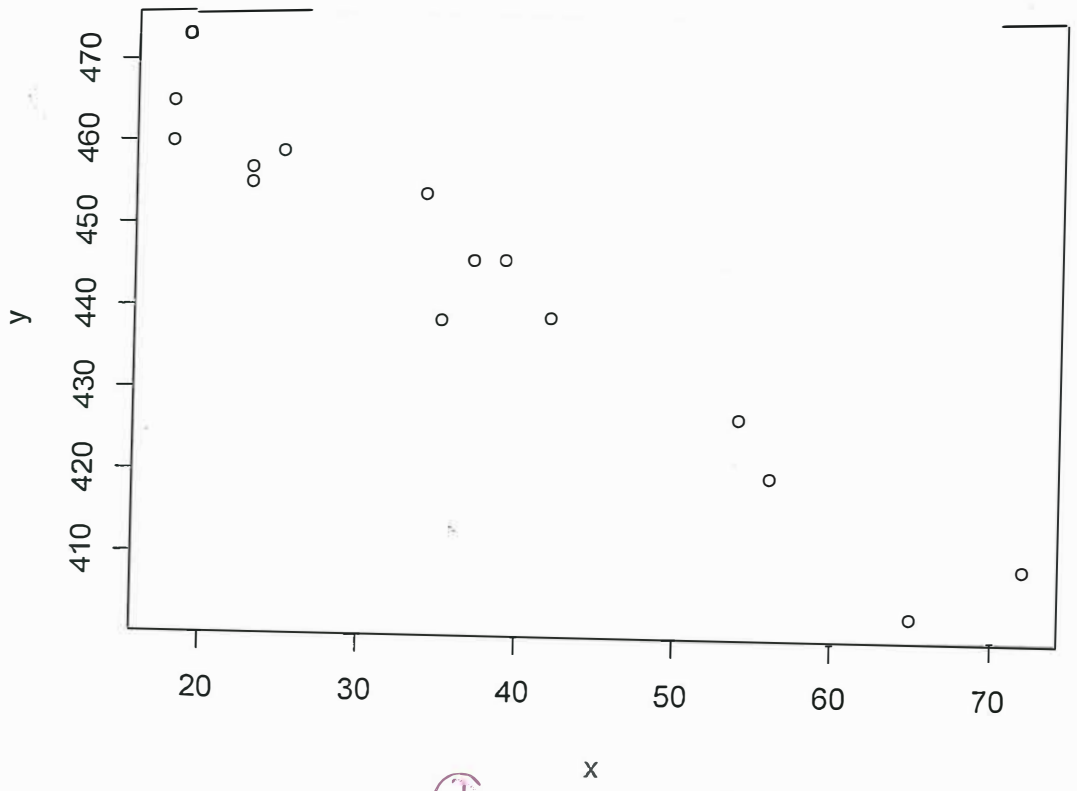
(c) For (i) the power is $1 - 0.3504 = \underline{0.6496}$ (1)

" (ii) " " " $1 - 0.1209 = \underline{0.8791}$ (1)

The test is more powerful as p increases. (1)

6.

(a)



There is a strong (1) negative (1) association between sleep and age.

(b) $S_{xx} = 25188 - \frac{1}{15} \times 560^2 = 4281.33$

$S_{xy} = 243444 - \frac{1}{15} \times 560 \times 6652 = -4897.33$ (1)

$\hat{b} = \frac{S_{xy}}{S_{xx}} = \frac{-4897.33}{4281.33} = -1.144.$ (1)

$\hat{a} = \bar{y} - \hat{b}\bar{x} = 443.467 + 1.144 \times 37.333$

$= 486.176$

Equation is $y = 486.176 - 1.144x$ (1)

(c)

Standard error of estimate at $x=60$: $\hat{\sigma}\sqrt{\frac{1}{15}+\frac{(60-37.333)^2}{4281.33}+1}$ (1)

$= 5.376 \times \sqrt{1.1867}$

$= 6.465$ (1)

95% C.I. is $(486.176 - 1.144 \times 60) \pm 2.160 \times 6.465$

$= 417.536 \pm 13.964$

$= (403.572, 431.5)$ (1)

(d) For $x=60$, the suggested equation gives

Sleep minutes $= 480 - \text{Age}$

$= 480 - 60$ (1)

$= 420$.

This is consistent with the confidence interval in (c) so the claim may be correct. (1)

(e) The confidence intervals calculated for individual values will be greater than those for sample means. 1 This is because of the Central Limit Theorem. 1